

CS 3081

5 Oct 2026

CASSIA

Workload, Energy, Heat & Water Optimization
Application: AI Data Center Supporting Hajj Services

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THE PROBLEM

AI data centers consume large amounts of energy and water to operate and cool their servers. In Saudi Arabia, producing cooling water through desalination also requires significant energy. During Hajj, AI services may experience high and changing workloads, which can increase the demand for both energy and cooling.

OUR SOLUTION

01



Solar Energy

02



AI Workloads

03



Heat Recovery

04



Desalination

05



Water Storage

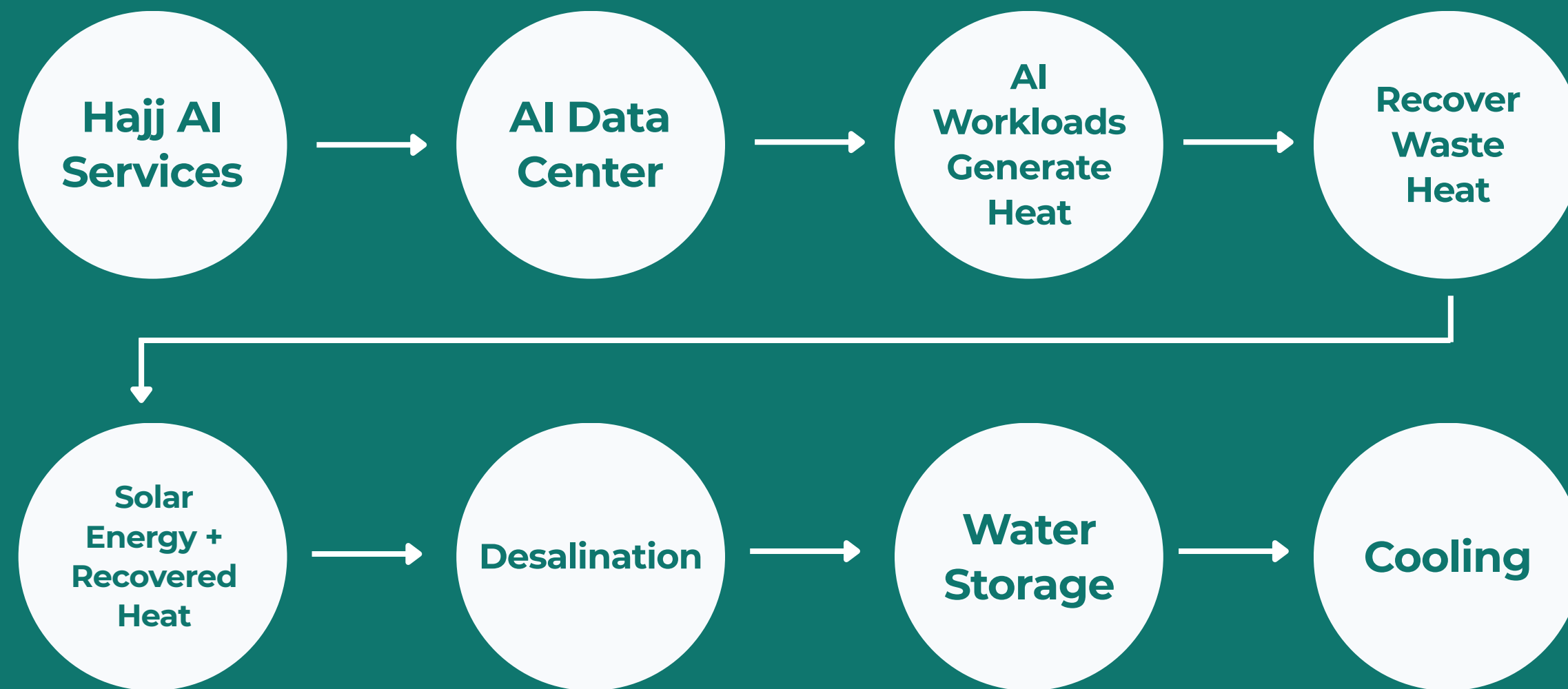
06



Cooling

We propose a simulated AI data center that combines solar energy, workload management, heat recovery, desalination, and water storage. An intelligent A* agent decides how to use these resources efficiently while keeping critical Hajj AI services running.

HOW IT WORKS



AI TECHNIQUES

A* Search

Used to find an efficient sequence of actions for managing:

▶ AI workloads

▶ Energy

▶ Heat recovery

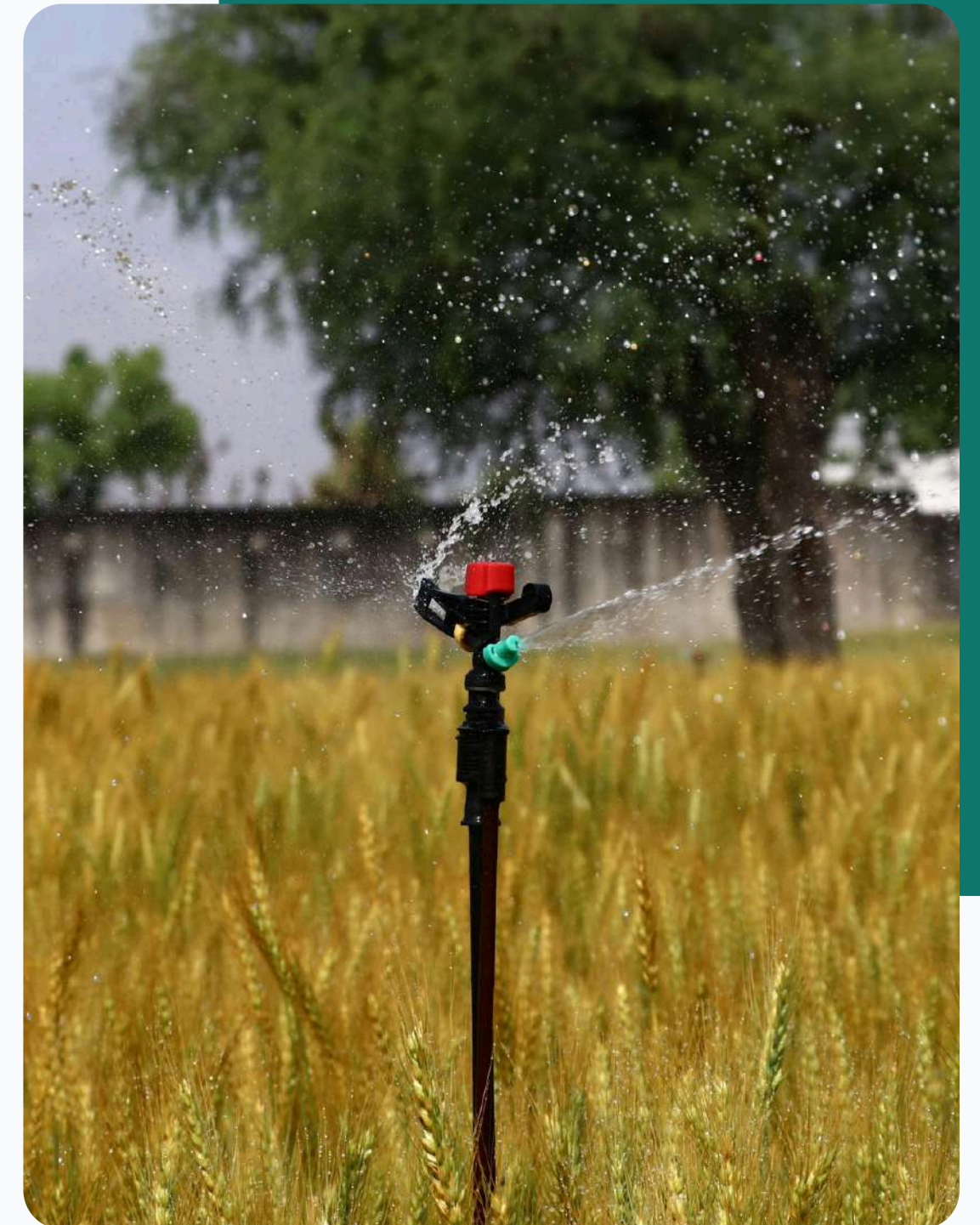
▶ Water production

▶ Cooling

Optimization

The agent aims to:

Reduce external water and energy use while keeping critical AI services running.



HAJJ SCENARIO

The data center can support AI services such as:



Crowd
monitoring



Fix leaks
immediately



Emergency
response

Some workloads are critical, while others can be delayed.
This allows the agent to prioritize important services
during periods of high demand.

CHALLENGES

The main challenge is balancing AI workload, energy, heat, and water. Solar energy and AI workloads can change over time, while the data center must still maintain enough cooling and keep critical Hajj services running. The system must make efficient decisions without wasting resources.



Managing
limited water
resources



Balancing energy
and water use



Handling changing
AI workloads



Maintaining
cooling
requirements



Keeping critical
services running

WHY IT MATTERS

The project aims to reduce the data center's dependence on external freshwater and grid energy. It also demonstrates how waste heat and renewable energy can be reused to support cooling, making AI data centers more sustainable.

EVOLUTION



Initial Idea

Water-Self-Sufficient AI Data Center



Added Resource Management

Workload + Energy + Water



Added Heat Recovery

Reuse server heat



Added Hajj Scenario

Apply the system to Hajj-related AI services



Final Idea

An A-based intelligent system* that manages workloads, energy, heat, and water to improve data center sustainability.



MAIN GOAL

Reduce external freshwater and energy use while maintaining the performance of critical AI services.

▶ Solar Energy

▶ AI Workloads

▶ Heat Recovery

▶ Desalination

▶ Water Storage + A Search*

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THANK YOU

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